DRAFT APPENDIX FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT*
HISTORICAL NOTE: INTEGRAL CAYLEY NUMBERS, 1923–1946

CLAUDE OPUS 4.7

Historical note: integral Cayley numbers, 1923–1946

The construction this paper builds on was developed across four primary papers between 1923 and 1946. Each was obtained in the original and re-derived in exact arithmetic in this project (`python_project/src/verify_{dickson_1923,kirmse_1924,mahler_1942,coxeter_1946}.py` and `verify_kirmse_1924_forensic.py`; verification notes in `paper/reviews/`), for Mahler's covering-radius theorem by exact-arithmetic sampling.

Dickson (1923) [Dickson1923] gave the first construction: the Cayley–Dickson doubled product and an explicit 8-element basis of an order O_D, with all 64 basis products closing. His Theorem XV states three maximal orders containing the eight basis units; the true count is *seven*. The cause is in his derivation (p. 321): he assumes the orders contain the Hurwitz quaternions, and exactly three of the seven do.

Kirmse (1924) [Kirmse1924] took up the problem independently. His multiplication table is itself a correct octonion algebra, and he derived a correct *necessary* form for the maximal orders, without proving sufficiency; thirty E_8-type lattices realize the form. Among the eight of those lattices that are stable under multiplication by units (an unstated narrowing of the criterion, in our reconstruction), he exhibited the one, J_1 (p. 70), that is not an order. The remaining seven of his eight are the genuine maximal orders.

Mahler (1942) [Mahler1942] developed the arithmetic of a maximal order: for every Cayley number X there is an integral G with $N(X-G) \le \frac{1}{2}$ (sharper than Dickson's 5/4), and every left or right ideal is principal, generated by an integer multiple of an integral Cayley number of norm 1, 2, or 4.

Coxeter (1946) [Coxeter1946] identifies Kirmse's error with an explicit nonclosure witness and records that the remedy is due to R. H. Bruck: "Bruck's domain J can be derived from J_1 by transposing two of the i's" (§4). This is a transposition of two imaginary basis units, a linear involution of $\mathbb{R}^8$; the $\binom{7}{2} = 21$ such transpositions yield the seven maximal orders in seven sets of three. His §§6–13 develop the $E_8/5_{21}$ geometry of the corrected order. A postscript (§14) records his after-the-fact awareness of Dickson and Mahler and identifies Dickson's undercount with exactly the cause above.

It is curious to find that the Coxeter–Bruck transposition would come in useful here: the same linear involution of $\mathbb{R}^8$, acting now on a different object. The Coxeter–Bruck transposition acts on a maximal-order candidate; the present σ acts on the sublattice Ls of Wilson's Leech construction.

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Drafted by the AI agent Claude Opus 4.7 (Anthropic) at the direction of Jens Köplinger, as part of the systematic investigation recorded in this repository (`prompt_logs/`). The version that appears in the main paper is owned by the principal investigator.

References

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[Kirmse1924] J. Kirmse, *Über die Darstellbarkeit natürlicher ganzer Zahlen als Summen von acht Quadraten ...*, Ber. Verh. Sächs. Akad. Wiss. Leipzig, Math.-Phys. Kl. **76** (1924), 63–82.

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